SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Computer Vision with OpenCV

COURSE CODE: DSA0202

COURSE OUTCOME COVERED

CO1: *Apply the fundamental concepts of image formation, pinhole to perspective camera models, homogeneous coordinates, and multi-view geometry and interpret real-world visual scenes using OpenCV or MATLAB. (BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 20 Total Marks:40 Annexure A

SHORT ANSWER TEST

Code of Conduct:

I K.Pujitha (192524149) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)	Marks
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding	

Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer	
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application	
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response	
Clarity & Presentation	**10**	Clear, concise, and well organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand	

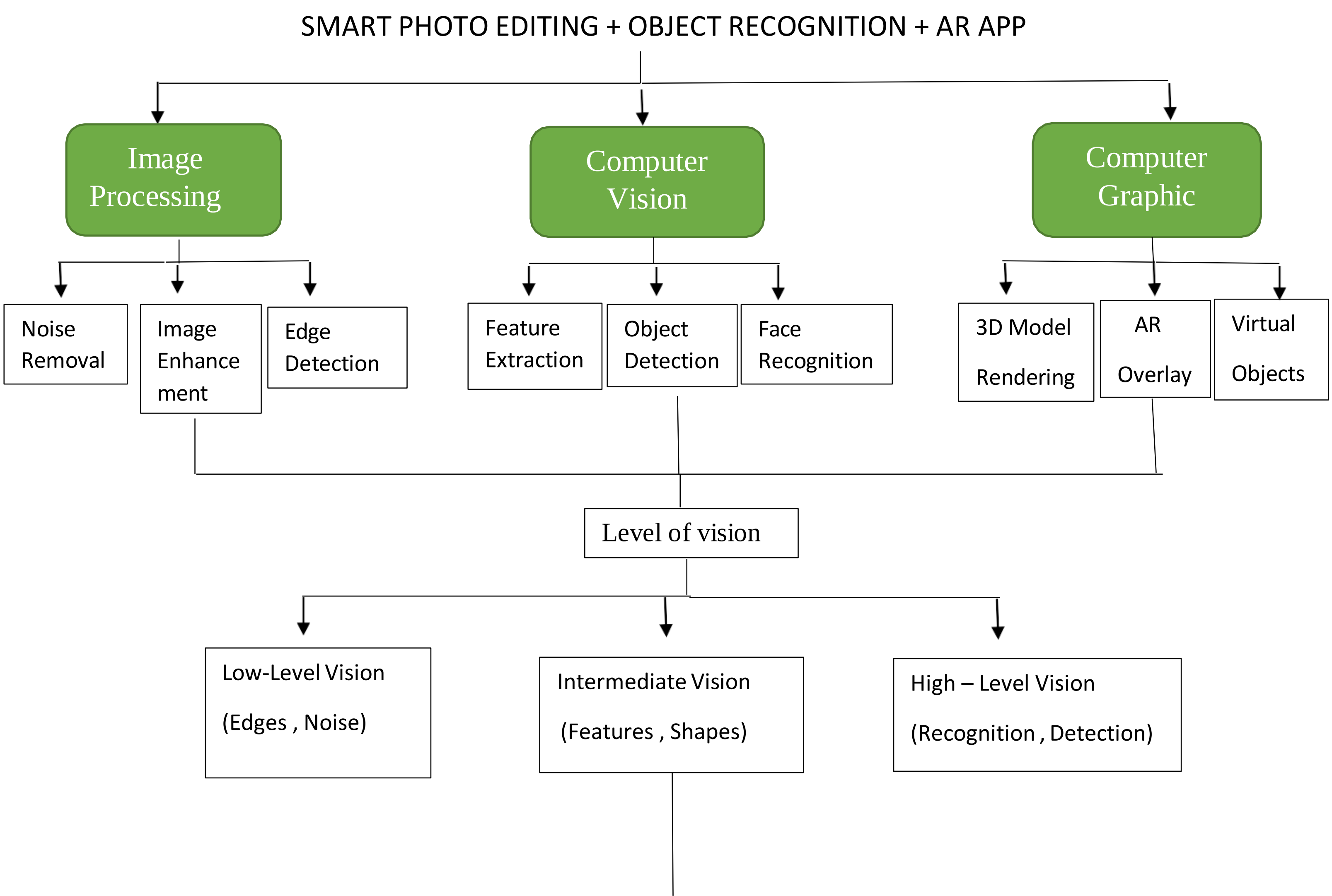
1. A company is developing a smart photo editing + object recognition + AR app.

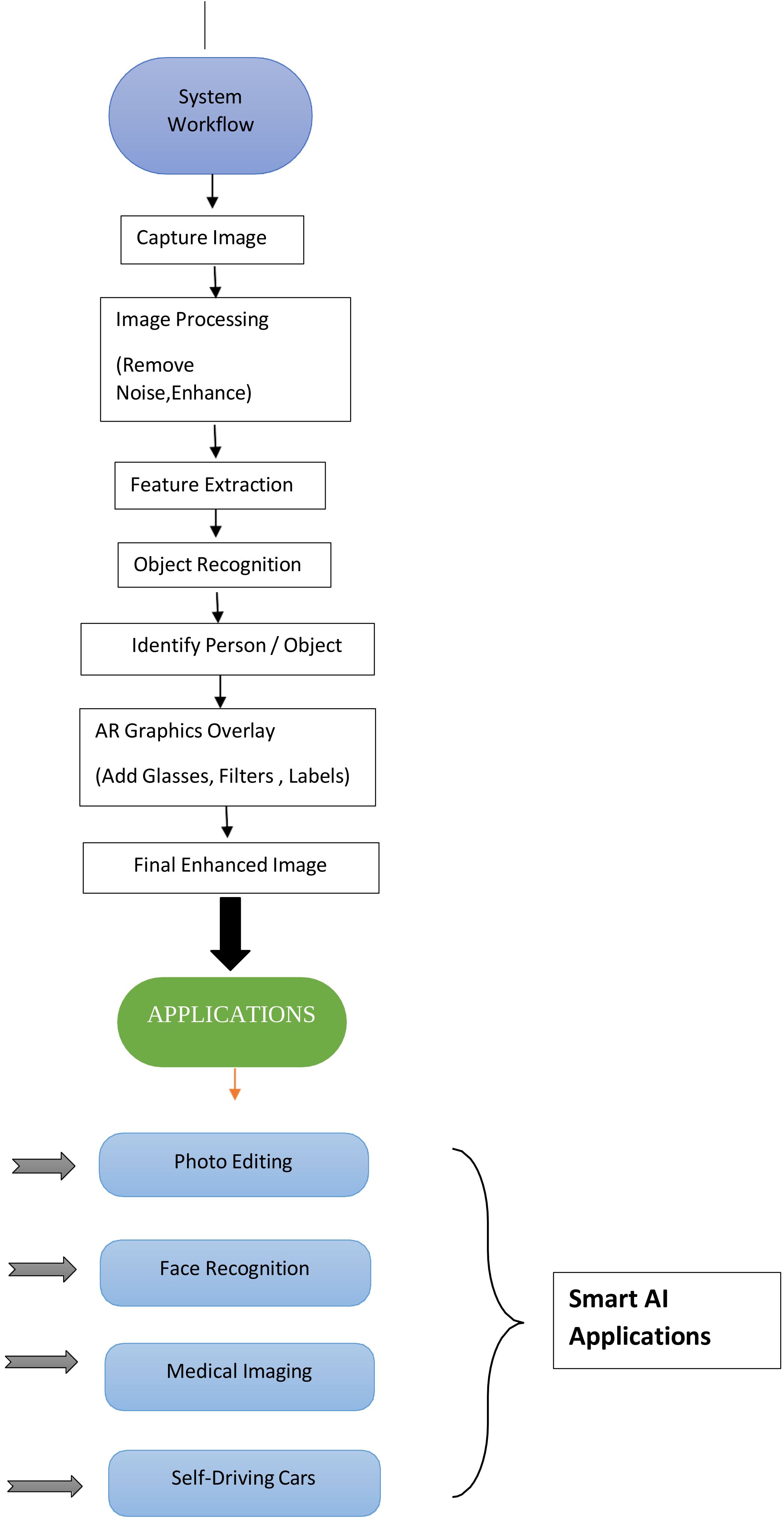
Create a concept map connecting:

- Image Processing
- Computer Vision
- Computer Graphics
- Levels of Vision
- Applications

Show:

- Workflow of the system
- Interconnections between components
- Real-world examples




System Workflow
Capture Image
Image Processing
(Remove Noise,Enhance)
Feature Extraction
Object Recognition
Identify Person / Object
AR Graphics Overlay
(Add Glasses, Filters , Labels)
Final Enhanced Image
APPLICATIONS
Photo Editing
Face Recognition
Medical Imaging
Self-Driving Cars
Smart AI Applications

Explanation

A Smart Photo Editing + Object Recognition + AR application combines Image Processing, Computer Vision, and Computer Graphics.

- **Image Processing** improves image quality by removing noise, enhancing brightness, and detecting edges.
- **Computer Vision** extracts features, recognizes objects, and identifies people using different **levels of vision**:
  - **Low-Level Vision:** Image enhancement and edge detection.
  - **Intermediate-Level Vision:** Feature extraction and object detection.
  - **High-Level Vision:** Object recognition and decision making.
- **Computer Graphics** generates virtual objects and overlays them on real-world images using **Augmented Reality (AR)**.

Workflow:

Capture Image → Image Processing → Feature Extraction → Object Recognition → AR Overlay → Final Enhanced Image

Real-world Examples:

- Snapchat and Instagram filters
- Google Lens
- Pokémon GO
- Medical image analysis
- Self-driving cars